

Solving the layout design problem by simulation-optimization approach—A case study on a sortation conveyor system

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ABSTRACT

Because of increasing demands in parcel transportation, it is becoming increasingly important for logistics distribution centers to save time and money while dealing with parcel sortation operations. The role played by facility and layout planning in logistic distribution centers has a great impact on the performance of parcel sortation. In this study, a simulation-optimization approach is adopted to solve the sortation conveyor layout problem, particularly the design of shortcuts in the sortation system that enhances processing capacity. The capacity is defined by the throughput of the sortation system. A genetic algorithm with local search is proposed to find the optimal design of the sortation system. Simulation is used to evaluate the performance of algorithm results. Moreover, an experimental design is applied to optimize the parameter setting of the algorithm and the system. A practical case is studied to demonstrate the applicability of the proposed approach.

1. Introduction

Facility layout design is a critical issue in both manufacturing and logistics systems [1–3]. Because of the growth of e-commerce, the parcel service industry has become one of the fastest-rising segments of logistics services [4]. The rapid growth of e-commerce has led to a significant increase of shipped parcels in recent years. Higher levels of customization, larger quantities, and smaller parcel batches have made parcel processing and delivery more complicated. However, to deliver the right parcel to the right place, the process of gathering parcels from different places and distributing them to the planned location also plays an important role in parcel service industry. This process is known as “sortation,” and its goal is to classify items according to their destination and move different items to the corresponding output ports.

With the recent advances in technology, automated sorting systems with sensors and conveyors have largely replaced manual sortation work and, subsequently, reduced cost and workload. Fig. 1 shows the automated system of the sorting process. When a parcel is placed on the conveyor, the sensor scans the zip code on the parcel and transports it to the output port of the corresponding destination. The system increases the efficiency of the sortation operation and reduces human effort. This automated operation had become a trend in not only the parcel service industry but also other applications that needed to transport items (e.g., baggage conveyor systems in airports). An automated sortation conveyor layout design is considered, and it is assumed that there is no manual operation needed in the conveyor system.

According to Drira et al. [5], layout configuration can be divided into the following categories by the shape of the layout: single

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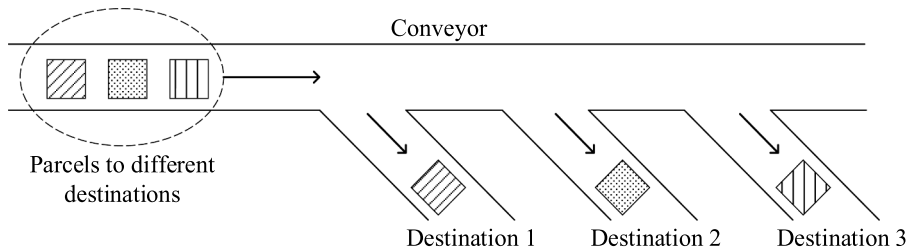


Fig. 1. Automated sorting process.

row, multirow, loop layout, open field, and multifloor. The loop-layout design problem (LLDP) arises when the machines are arranged in a closed ringlike network and the materials are transported around this network in only one direction [6]. In this study, loop layout is taken as the initial design.

The aim of this paper is to use a genetic algorithm (GA) to solve the conveyor layout design problem (CLDP) with a shortcut in a sortation system. A simulation model using Plant Simulation is utilized to evaluate the performance of the system with different sets of parameters. The objective is to maximize the capacity of the loop sortation conveyor system. The capacity in this study is defined by the throughput of the sortation system, which means the number of sorted parcels in specified time. The solution proposed via GA is compared with the solution of a greedy algorithm. Local search is further applied to improve the performance of the GA, which is called “local-search genetic algorithm” (LSGA) in the following content. Computational results demonstrate that the proposed LSGA can achieve better results than the greedy algorithm. The contribution of this paper is twofold from both academy and practical perspectives. From the academy perspective, the genetic algorithm with local search is proposed to find the optimal shortcut design of the sortation system. In particular, most of the earlier works did not consider shortcut in such a sortation system. From the practical perspective, this paper utilizes a practical case to validate the applicability of the proposed approach.

2. Literature review

2.1. Conveyor layout design problem (CLDP)

Conveyor systems have been studied for the past 30 years as a means of storing and transporting units and have become an integral part of many services, manufacturing, and distribution facilities [7]. Sonderman [8] studied a conveyor system with a single loading and unloading station with stochastic arrivals at the loading station. Bastani [9] considered a closed-loop conveyor system having a single loading and multiple unloading stations that was a similar system to that of Sonderman [8]. Hong et al. [10] studied several heuristics to optimize the location of crossovers in a conveyor-based automated material-handling system (AMHS) for a semiconductor wafer fabrication facility. Zhou et al. [11] proposed an analytical model for a spine layout of continuous flow with conveyors and crossovers in wafer fabrication systems.

The problem of sorting system design and control has been discussed in recent years within logistics industries. Bozer and Hsieh [12] developed two analytical models to approximate the expected waiting times at loading stations in a closed-loop, unidirectional conveyor system with fixed-size windows. The results can be used for estimating the expected work in process (WIP) at loading stations. Wang and Zhou [13] studied the characteristics of a conveyor sortation system, developed a generic model of a complex conveyor network, and used the delay and stock abstraction to develop an approximate analytical method using sample path analysis and a dynamic network flow model. Werners and Wülfing [14] proposed a model that minimized the necessary manual transportation effort by layout modifications. The achieved reduction of manual transportation effort led to a significant increase in runtime quality, a decrease in costs, and balanced utilization of the sorting and distribution unit. Fedtke and Boysen [15] evaluated the impact of parallel conveyors, multiple loading stations, and different loading rules on different design alternatives of closed-loop tilt tray sortation conveyors applied in the parcel service industry. Furthermore, a novel destination assignment problem was defined. Phoon et al. [16] used virtual reality technology for loop layout planning to reduce the gap between numerical results and the real situation through an enhanced human-machine interface. A virtual loop layout model has been developed as an intuitive and interactive platform for loop layout planning and evaluation in real-time control. According to the abovementioned studies [7–16], design of shortcut was seldom considered in the conveyor layout design problem. Hence, a valuable study issue emerges to optimize the design of shortcut to meet practical needs.

2.2. Simulation-optimization

Simulation-optimization can be defined as the process of finding the best input variables from all possibilities without explicitly evaluating each possibility. The objective of simulation-optimization is to minimize the resources spent while maximizing the information obtained in a simulation experiment [17–19]. Nowadays, simulation-optimization is widely adopted to analyze the behaviors of complex systems, because it is capable of solving diverse practical problems that involve randomness. Chávez-García and Castillo-Villar [17] proposed a simulation-based method for setting parameters in metal-cutting operations. Fernández-Rodríguez et al. [18] designed an on-line calculation of the eco-driving of a high speed train by means of a dynamic multi-objective optimization

algorithm. Ozdemir and Kumral [19] adopted a simulation-based optimization of truck-shovel material handling systems in multi-pit surface mines.

In the logistics field, many researchers used the simulation-optimization approach. Layeb et al. [20] applied a simulation-optimization approach in stochastic freight transportation for solving the scheduling problem. Hou [21] developed a simulation model with FlexSim software and experimented with the model to observe the effect different scenarios may have on the behavior of the model. The simulation results provided helpful information for decision making in the distribution center inventory strategy and warehouse scale. Clausen et al. [22] presented an approach that closely linked mathematical optimization and discrete-event simulation to improve parcel trans-shipment operations. However, for a layout design problem, Azadivar and Wang [23] considered dynamic characteristics and operational constraints of the manufacturing system and used the GA and simulation to solve the facility layout design problem. They applied the GA to optimize the layout for manufacturing effectiveness, while the simulation served as a system performance evaluation tool. Since the earlier works [17–23] have successfully introduced simulation-optimization to the real-world cases, this paper intends to adopt such an approach for the new problem of shortcut design.

2.3. Genetic algorithm (GA)

The GA is a metaheuristic based on the mechanics of natural selection and natural genetics [24]. The strength of GAs lies in the ability to search large spaces effectively and efficiently, which is useful for solving inherently intractable problems, known as NP-hard problems. The GA is considered superior to other metaheuristics, and this is especially true when it is applied to facility layout problems [23]. Tam [25] used the GA to solve function optimization problems and represented solutions as strings of finite length. The results were compared with a local search technique and demonstrated that a GA can be a viable tool for solving large-scale layout problems. Islier [26] studied a GA-based model for facility layout and obtained better-quality layouts by this method on the test problems available in the literature. The results showed the potential of the GA in solving complex facility layout problems.

In recent years, the use of GAs to solve facility layout problems has gained popularity among researchers. Aytug et al. [27] provided an overview on the use of a GA to solve operations problems. Reviewed papers were classified according to the problems they solve, including facility layout design, production planning and control, and line balancing. El-Baz [28] proposed a GA to solve the problem of optimal facilities layout in manufacturing systems design and was able to minimize the material-handling costs. Shayan and Chittilappilly [29] provided design, development, and experimentation results of a new GA. This approach produced legal chromosomes without any need for repairing procedures and required little processing time, even for solving test problems of 100 facilities. Raman et al. [30] implemented a GA to solve the layout problem modeled with queuing theory. Che et al. [31] studied a bi-objective optimization for multi-floor facility layout problem. This combination provided a unique opportunity to consider stochastic variations while achieving a good layout. The result minimized four parameters: WIP cost, material-handling cost, deviation cost, and relocation cost. Recently, Chen et al. [32] applied simulation-optimization approach to solve the sortation conveyor layout design problem. However, this previous work [32] did not consider local search to find the most acceptable performance in different cases. Moreover, response surface methodology to evaluate the optimal GA parameters was not included. Response surface methodology (RSM) is widely applied in simulation optimization. In particular, RSM is used in conjunction with GA to optimization problem. Kurtaran and Erzurumlu [33] studied the warpage optimization by RSM and GA. Dang [34] proposed general frameworks for optimization of plastic injection molding process parameters with GA and RSM. Li et al. [35] combined RSM and GA approach to optimize the fiber-reinforced composite injection molding process. Generally speaking, RSM is a metamodeling technique to express relationship between input and output in the form of quadratic polynomial [34]. That is, RSM estimates local gradients to search for the optimal parameter. Such gradients are on the basis of local first-order polynomial approximations [36]. Therefore, RSM also can be adopted to search the optimal GA parameters, such as in [37]. To apply GA with local search to the new problem of shortcut design, the encoding of genes, fitness function, and mechanism of local search are further introduced in this paper.

3. Problem definition

This study focused on CLDP with a shortcut, and applied an LSGA to overcome the shortcut location decision problem. Because there is little research focusing on the shortcut layout design in sortation systems based on CLDP, this research develops a simulation model and the LSGA to evaluate shortcut performance.

3.1. Alternative design of loop sortation conveyor

Fedtko and Boysen [15] summarized different layout alternatives for closed-loop sortation conveyors. The system included tilt trays on the conveyor that could load and transport parcels. When reaching a corresponding outbound conveyor, the tray tilts, and the parcel slips to the chutes. Two layout alternatives related to this study are summarized below.

- (a) Single loading station: only one loading station is available. Each tray can be loaded with a parcel (at most) once per cycle of the conveyor.
- (b) Multiple loading stations: compared with a single loading station, additional loading stations could provide multiple tray loads per cycle. In the best-case scenario, each tray of a conveyor with n segments can be loaded n times per cycle. However, the worst case would have a parcel traveling all along the conveyor, only to be tilted just before reaching its loading station again and, hence, still only achieving a single load per cycle.

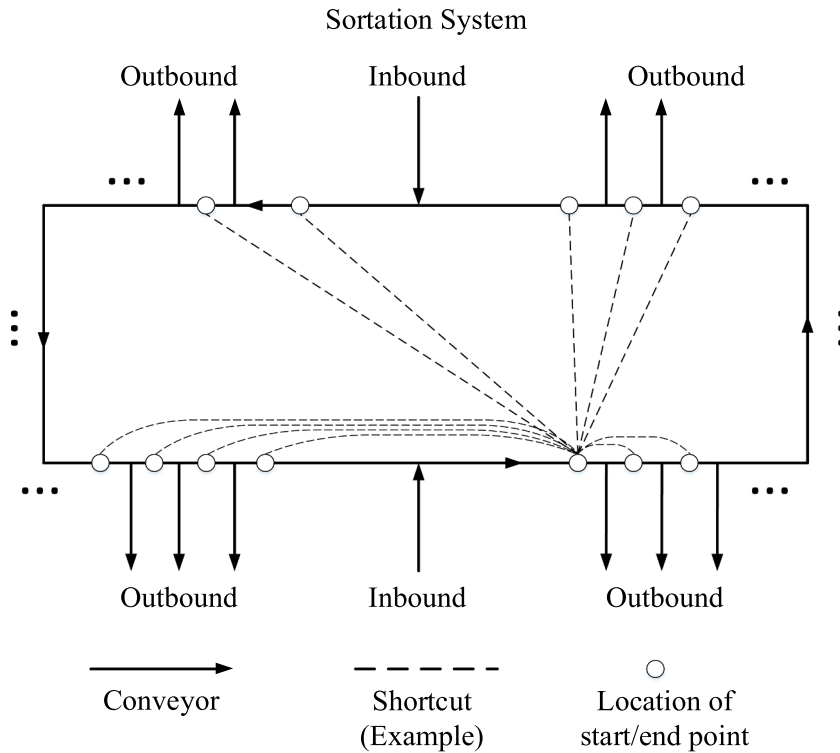


Fig. 2. The example of loop conveyor.

For this study, the objective is to maximize the capacity of the sorting system. Therefore, a multiple loading station is chosen. Shortcuts are added into the system to enhance the capacity of systems. Two loading stations are used, and there is no division of the conveyor into inbound and outbound segments. This means that not all parcels are unloaded before reaching the next loading station. Shortcut conveyors are used to enhance the capacity of the sorting system. The flow direction is unidirectional.

3.2. Problem statement

Consider a loop conveyor as shown in Fig. 2. The processes before parcels are loaded onto inbound conveyors and unloaded from outbound conveyors are outside the scope of this study. To increase the throughput of the sortation system, two inbound conveyors are set in two opposite sides of the loop conveyor. Several outbound conveyors are set, representing different destinations for the parcels. Outbound conveyors are set evenly along the loop conveyor. The flow of the conveyor is counterclockwise.

Shortcuts are extra routes of the loop sortation system to transport parcels faster. Although more shortcuts imply a better capacity of the system, when considering the number of shortcuts to set in the system, the related costs, such as facility cost, manpower, and maintenance, need to be evaluated. There are I potential locations for start/end points in Fig. 2. Dashed lines in Fig. 2 depicts an example of a shortcut conveyor. When a parcel reaches the loop conveyor and the corresponding outbound is on the opposite of the loop, a shortcut can help to transport the parcel to the opposite side faster. This also increases the utilization of the loop conveyor by reducing the amount of occupied space. Please note that Fig. 2 is depicted in general form and it can be applied to any other sortation conveyor system.

3.3. Assumptions and notations

Before starting with the problem analysis, the assumptions made are listed below.

- 1 The destination of every parcel and the allocated destination of each outbound conveyor are known.
- 2 The parcel arrival interval follows a Poisson distribution.
- 3 The length and speed of the loop conveyor are fixed.
- 4 The size and percentage of all types of parcel are known.
- 5 The numbers of parcel types, inbound conveyors, and outbound conveyors are fixed.
- 6 The time the system takes to scan the zip code of the parcel is negligible.

The notations used in the proposed mathematical model are defined in Table 1.

Table 1
Notation list.

Indices	
I	Total amount of locations for start points of shortcuts
i	Index of start point of shortcut, where $i = 1, 2, \dots, I$
J	Total amount of locations for endpoints of shortcuts
j	Index of end point of shortcut, where $j = 1, 2, \dots, J$
Parameter	
S	Required amount of the shortcuts
Decision variable	
λ_{ij}	1: shortcut λ is designed from start point i to end point j 0: otherwise
λ	Matrix of all designed shortcuts $\lambda = (\lambda_{ij})_{I \times J}$

3.4. Problem formulation

The objective is to optimize the capacity of the system, which is the sum of the outputs of all outbounds per hour, denoted $TH(\lambda, \omega)$. Eq. (1) presents the objective.

Objective function

$$\text{Maximum} E[TH(\lambda, \omega)] \quad (1)$$

Constraints

$$\sum_i^I \sum_j^J \lambda_{ij} = S \quad \forall i \neq j \quad (2)$$

$$\lambda_{ij} \in \{0, 1\} \quad \forall i \neq j \quad (3)$$

Objective function (1) is to optimize the expected throughput of the sortation system in a simulation model. In Objective function (1), ω stands for the parameter of parcel arrival interval, which follows Poisson distribution. Eq. (2) ensures the number of available shortcuts. Index i demonstrates the start point ($i = 1, 2, 3, \dots, I$), where index j is the end point of the shortcut ($j = 1, 2, 3, \dots, J$). Note that, the start point and endpoint of a shortcut are selected among the given location. Besides, the start point and endpoint are not in the same locations. To ensure that the start point and endpoint are not in the same location, it is necessary to denote that $i \neq j$ in Eqs. (2) and (3). In Eq. (3), the internal flows with a specific conveyor direction are represented by λ_{ij} where the parcel travels from location i to location j . For example, $\lambda_{ij} = 1$ if a shortcut is set between location i (start point) and location j (endpoint). In the other side, $\lambda_{ij} = 0$ if there is no shortcut between location i and location j .

In this study, three test cases with different numbers of chosen shortcuts are considered. Test cases 1, 2, and 3 have two, four, and six shortcuts, respectively. For example, if connect point $I = 12$, for test case 1 with two shortcuts, there are $2 \times C_2^{12}$ shortcuts that can be chosen. The total number of possible solutions will be $C_2^{2 \times C_2^{12}} = 8646$. For test case 2 with four shortcuts, the total number of possible solutions is 12,082,785. Last, for test case 3 with six shortcuts, the total amount of possible solutions is 6,547,258,432. When the number of chosen shortcuts increases, it is nearly impossible to try every possible solution and evaluate the performance of the solutions. Consequently, it is essential to develop a heuristic algorithm to solve the problem.

4. The proposed approach

To solve the CLDP in this study, simulation and optimization are applied to find the best solution for the problem. In the simulation phase, a simulation model is constructed to simulate the operation and the throughput of the sortation system with predefined assumptions. In the optimization phase, the GA is adopted to find the optimal solution from all the possible solutions generated in the simulation phase. In Section 4.1, the simulation model constructed for the proposed problem is presented. Section 4.2 covers the process of adopting the GA to find the optimization solution. The procedure of the proposed simulation-optimization approach is illustrated in Fig. 3. The interplay between GA (optimization) and simulation can be described according to Fig. 3. We can find that the initial population is generated first. Subsequently, the evaluation is processed by simulation approach to obtain the throughput as a fitness value. In other words, the simulation phase can be seen as a fitness function. Furthermore, based on the obtained throughput, crossover, mutation, and local search are further processed by GA.

4.1. Simulation modeling

To design the simulation model, a practical sortation system framework using Plant Simulation [38], a 3D dynamic simulation software, is employed. There are three areas to be constructed in the simulation model: (i) objects and layout of the system, (ii) commands of operation rules behind the system, and (iii) analysis object embedded in the simulation model.

First, the layout of the sortation system was converted into the corresponding objects in simulation model. In the simulation environment, source of parcel was modeled by the object "Source" and was placed on the inbound conveyors. The temporary storage

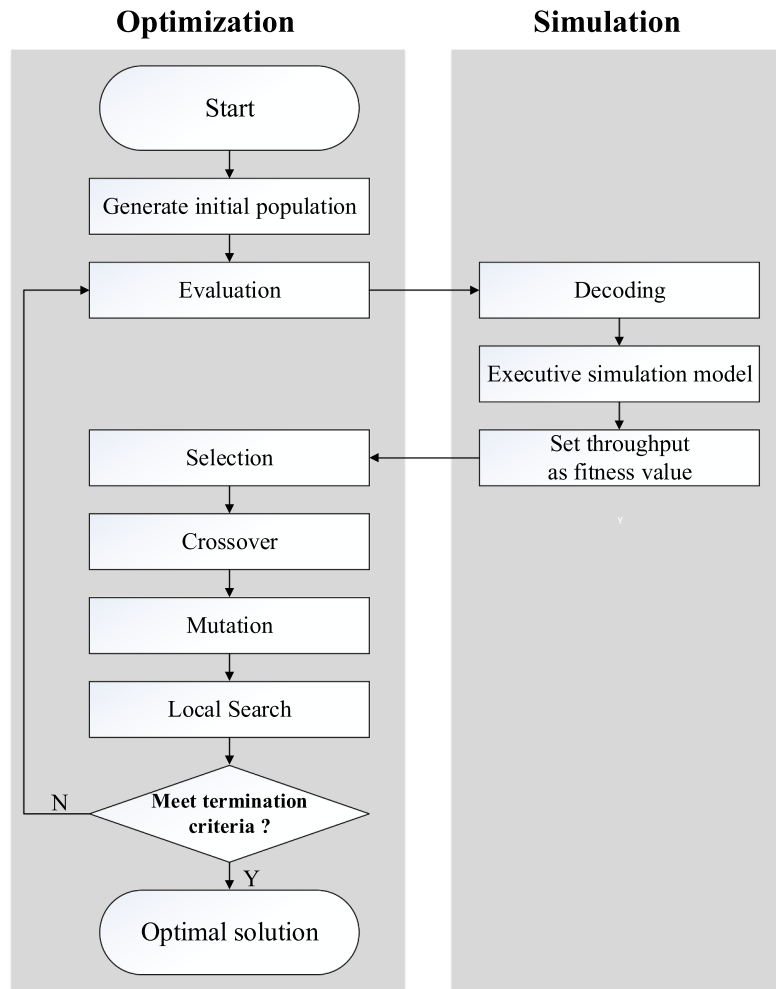


Fig. 3. The proposed simulation-optimization approach.

areas were modeled by the object “Store”, and is placed on the outbound conveyors. “Line” is utilized to represent the conveyor that connected all the objects, simulating the flow of the conveyor.

Second, the predefined parameter for this study was configured into the corresponding objects in the simulation. After the layout is complete, process design and data generation needs to be programmed. Simtalk, the scripting language in Plant Simulation, was used to change the behavior of the objects. Due to the stochastic characteristic of source, repetition of the simulation needs to be performed and their results recorded.

Finally, after finishing the loop layout, the next step was to construct the shortcuts and define and related rules. Simtalk is further used to construct the shortcuts and connectors iteratively. The procedure of simulation modeling is as follows.

4.1.1. Facility layout

For the purpose of validating the applicability of this research, a real-world conveyor sortation system was studied as a demonstration. The modified layout of loop sortation system is shown in Fig. 4. The layout includes two inbound conveyors, the loop conveyor, and 10 outbound conveyors. As a parcel enters the system, it is loaded onto one of the inbound conveyors and then transported to loop conveyor and sorted to one of the outbound conveyors according to its destination. Because the main problem this study attempted to address was the throughput of the system, the operations before loading to inbound conveyor and after sortation were not considered in the simulation model.

The objective of this study was to find the best design for shortcuts that enhances the capacity of the system. The key performance indicator for the performance evaluation in the simulation model was the throughput of the loop sortation system over a period of one hour, which can also be viewed as the sum of all sorted parcels after one hour.

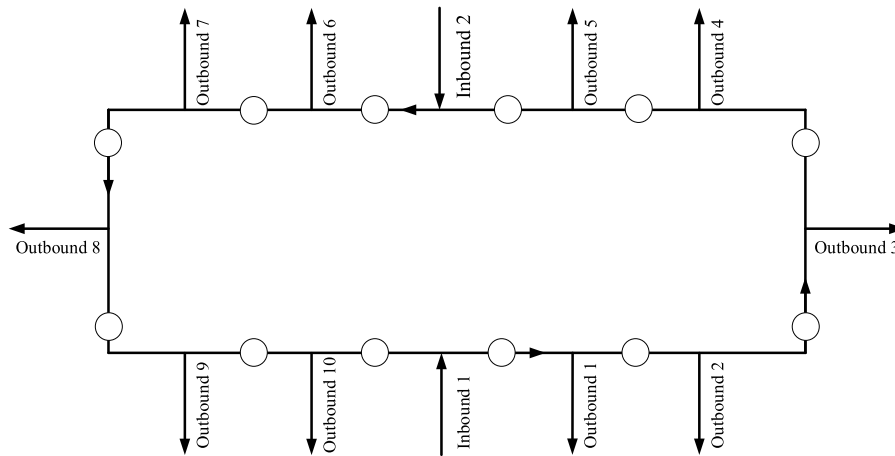


Fig. 4. Loop sortation layout.

Table 2

The information of parcel type and outbound conveyor.

Parcel type	Corresponding outbound conveyor	Parcel length	Relative frequency
A	1	0.5 m	10%
B	2	0.5 m	10%
C	3	0.5 m	10%
D	4	0.5 m	10%
E	5	0.5 m	10%
F	6	0.5 m	10%
G	7	0.5 m	10%
H	8	0.5 m	10%
I	9	0.5 m	10%
J	10	0.5 m	10%

4.1.2. Simulation setting

Table 2 shows the relationship between parcel type and outbound conveyor. There were 10 types of parcel, each with its own corresponding outbound conveyor (i.e., each outbound conveyor only served one type of parcel). The parcel arrival interval follows a Poisson distribution. After collecting real data from the postal service industry, the distribution of the parcel arrival interval was identified using the commercial software, Expert Fit. The parcel arrival interval follows a Poisson distribution with mean value 0.1. This means one parcel will arrive within ten seconds. In this case, each parcel was put into a single-size container no matter what size and shape the parcel is. Hence, the length of the parcel was fixed at 0.5 meter. The relative frequency was how often parcel types entered the system divided by all parcel types. The relative frequency of each parcel type was fixed at 10%, which means that the number of parcels for each outbound conveyor was balanced.

It was assumed that every conveyor within the system travels at a fixed speed of 2 m/s. Hence, the capacity of shortcut is 2 m/s as well. Since the shortcut is able to shorten the length between two locations, the throughput of the system would be increased. This includes the inbound conveyors, loop conveyor, shortcuts, and the outbound conveyors. In a real-world system, there are sensors on the conveyor that scan the zip code, which determines the destination of the parcels. The scanning process usually takes very little time, which means it is unnecessary for the conveyor to slow down. Therefore, the scanners were omitted in the simulation model, because they do not influence the capacity of the system.

The mechanisms applied for a parcel to enter or exit in a shortcut are further introduced as follows: For the route to choose, when a parcel meet a crossroads where there are more than one way to choose, there are three rules, which are described below.

- 1 Only the shortcut whose endpoint is before the parcel's destination can be chosen.
- 2 If there are any available shortcuts, transfer to one of the shortcuts, effectively placing a higher priority on taking shortcuts as opposed to staying on the loop conveyor.
- 3 If there are more than one available shortcut to choose from, the shortcut whose endpoint is closer to the destination of the parcel is chosen.

4.1.3. Verification and validation

The objective of the validation was to check the accuracy of the simulation model's representation for the real-world system. To evaluate the throughput based on the same situation, the original loop sortation system without any shortcut was adopted. Two inbound conveyors were included. In this study, throughput was compared by collecting one hour of data from both the simulation

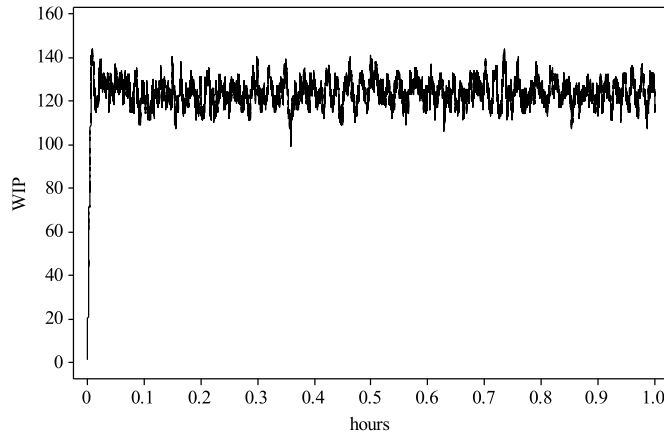


Fig. 5. The steady-state test of WIP vs. hour.

model and the real simulation plant. Then, the average of throughput was evaluated for calculation of absolute relative error (ARE) value between the output of the simulation model and manufacturing plant. The average throughput of the simulation model per hour was 19,113 and of the distribution center was 19,512. The ARE value was 2.045%, which is located in the acceptable range. Hence, the simulation model is valid. After the step of verification and validation, the simulation model was kept for further analysis.

Before moving on to the optimization phase, the steady-state of the simulation model also had to be evaluated to ensure that the sortation system in the simulation model was stable for further analysis. The simulation model was set to run for one hour and the generated data were collected. The changes in WIP quantity of the loop conveyor were observed to figure out the steady-state of the system. Fig. 5 shows the steady-state test of the WIP amount in the loop conveyor during the one hour simulation run. The average steady time was 0.0125 h (equal to 45 s). The mean number of WIPs on the loop conveyor was 124.3 with a standard deviation of 7.324.

4.2. Genetic algorithm

The GA was applied in the optimization phase in this study to solve the shortcut layout design problem, with the objective of maximizing system throughput. The process is documented in the sections below.

4.2.1. Initial population

Generating the initial population is the first step of the traditional GA. In the proposed GA, one possible solution is represented by a string that is known as a chromosome. Each random binary bit of 0 or 1 on the string represents a gene on a chromosome, which subsequently represents a shortcut; Fig. 6 demonstrates the representation of chromosomes, where 0 represents that the shortcut is not chosen, and 1 means the shortcut chosen. However, because the number of chosen shortcuts is predefined by test cases, the number of genes presented in one chromosome is the same as the number of shortcuts. The diversified initial solutions can help to get better performance in heuristics approaches.

4.2.2. Evaluation of fitness value

A fitness function was used to identify the performance of the solution and the parameters set. The objective of the sortation layout design problem of this research was to maximize the throughput of the sortation system. The fitness value (FV) for this study was obtained by the throughput of system:

$$FV = E(\text{Throughput of System}) \quad (4)$$

A higher fitness value represents a better solution and, hence, a higher likelihood for the solution to be preserved for the next generation. The detailed procedure of preserving better genes for the next generation is introduced in the Selection and Crossover sections.

λ_{11}	λ_{12}	λ_{13}	λ_{14}	λ_{15}	λ_{16}		λ_{ij}
0	1	0	0	1	1	...	0

$2 \times C_2^{12}$ chromosomes

Fig. 6. Chromosomes illustration.

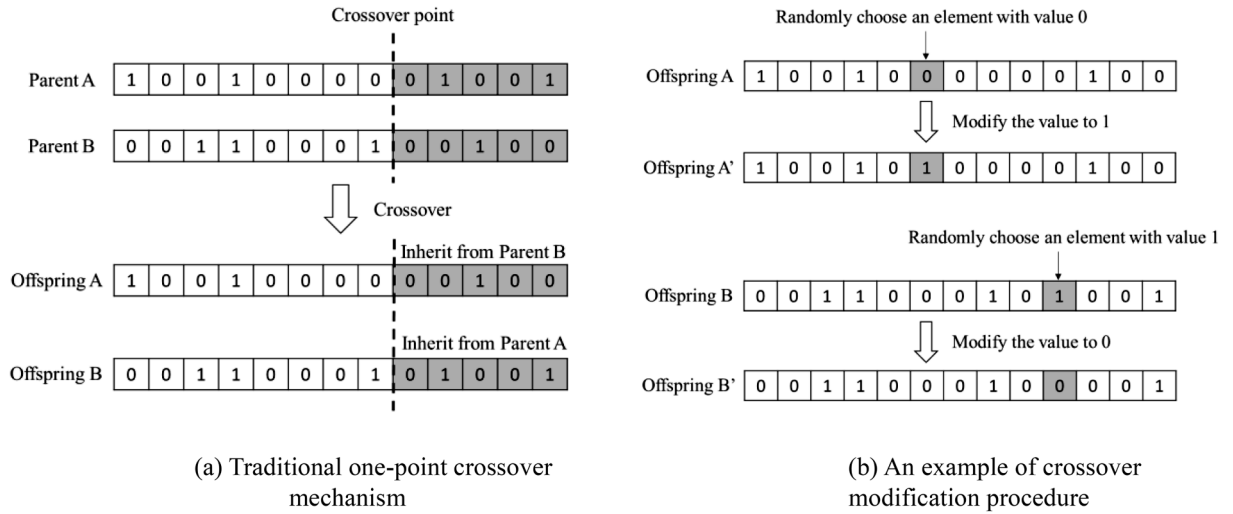


Fig. 7. Crossover procedure.

4.2.3. Selection

Selection is a process of selecting individuals for reproduction. The two most well-known methods are roulette wheel and tournament selection. In the roulette wheel method, selection of parents is in accordance with the probability distribution of the revised fitness values [24]. Tournament selection includes running some tournaments among a few randomly chosen individuals from the population and selects the winner [39].

To ensure the fitness of the next generation, elite preservation is conducted during this step. The fitness value of the entire population N is evaluated, and the top $N/2$ chromosome is preserved for the crossover phase. Then, the rest of the chromosomes for the mating pool are selected by a modified roulette wheel method. This method replicates nature in that chromosomes with better fitness values have a better probability of survival and go forward to form the mating pool for the next generation. However, weaker chromosomes are not without a chance. The rest of $N/2$ chromosomes are selected via the modified roulette wheel method.

4.2.4. Crossover

Crossover is an essential part of the GA. It simulates genetic recombination via breeding. The crossover phase mates two chromosomes to produce the next generation that shares the parents' genes. The more crossovers that take place, the faster the solution converges. In this study, pairs of mating individuals are randomly selected, and a probability pc , called the crossover probability, is determined. A uniform random number r is then generated, and, if $r \leq pc$, the two randomly selected individuals undergo recombination. Otherwise, if $r > pc$, the two offspring are simply copies of their parents.

In this study, the traditional one-point crossover combined with a modification process is used to meet the constraints on the shortcut quantity. In the traditional one-point crossover process, two parents (A and B) are selected using the Roulette wheel method in the selection phase, and one crossover point is identified. Fig. 7(a) shows an example of traditional one-point crossover.

However, the shortcut quantity constraint must be met to ensure that the solutions are valid. With regard to Fig. 7(a), if the shortcut number is assumed to be four, then offspring A and offspring B are invalid. Therefore, modification needs to be done to ensure the crossover makes sense. If the amount of value 1 is not as much as that of offspring A, then one element with value 0 will be randomly chosen and modified to value 1. However, if the amount of value 1 exceeds the constraint of offspring B, then one element with value 1 will be randomly chosen and modified to value 0. An example of the crossover modification procedure is depicted in Fig. 7(b). After making sure that all the offspring meet the required constraint, the next step is mutation.

4.2.5. Mutation

Mutation operation is used to increase the diversity of the population and prevent a local optimal solution. The steps for mutation are described below, and an example is shown in Fig. 8. The following steps are the detailed procedure of the mutation.

- Step 1: Generate a random number $r \sim U(0,1)$
- Step 2: If $r \leq pm$ (Mutation rate), randomly swap the two elements whose values are 0 and 1

4.2.6. Local search

In some circumstances, traditional GAs cannot find the global optimal solution. To improve the chance of finding a better solution, some researchers add local search mutation, which prevents the GA from being trapped in local optimal solutions. First, the best chromosomes of the parent generation are selected. Second, the process of pairwise interchange mutation is conducted on the selected chromosomes. In the end, a random number is generated. The steps in the local search phase are as follows. An example to

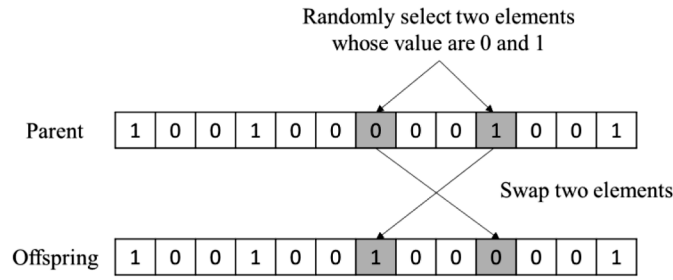


Fig. 8. Mutation procedure.

show the steps is provided in Fig. 9.

- Step 1: Choose a solution with best fitness value in parent generation
- Step 2: Select a shortcut and record its start point and endpoint
- Step 3: Generate a binary value (0 or 1) for r and s
- Step 4: If $r = 1$, then modify the start/end point (if $r = 0$, then do not modify)
 - If $s = 0$: start/end point -1
 - If $s = 1$: start/end point $+1$
- Step 5: Repeat step 2 and step 4 until finish modifying the rest of shortcut's start/end point of that best solution
- Step 6: Encoding the new solution to chromosome
- Step 7: Repeat step 2 to step 6 until modify the 10% of population size chromosome

5. Case evaluation

5.1. GA parameter setting experimental design

For the experiment, local search was removed temporarily to prevent interference. Before starting with the design of the experiment, the number of generations and some system parameters should be decided. First, three test cases were run to observe convergence. The convergence result of the three cases is summarized in Fig. 10. Because the growth of the obtained fitness value appeared sluggish before the 100th generation, the number of generations for experiments was set to be 100. The program was executed on a personal computer with Intel i7-8665 U CPU 1.90/2.11 GHz and 32 GB RAM. The computational time to completed an experiments was about 3.5 h (12,653 s).

A response surface methodology (RSM) design with three factors was implemented to evaluate the optimal GA parameter setting in the control variables experiments. Central composite design was used in the RSM to analyze the factors. After the best GA parameters were obtained from the RSM, they were substituted into the three test cases. After deriving the best GA parameters via RSM design experiment, these parameters are substituted into the three test cases. Local search is then added in the procedure as introduced in Section 4.2.6. The factors were population size (A), crossover rate (B), and mutation rate (C). The high level and low level of each factor were (A) [20, 100], (B) [0, 1], and (C) [0, 0.2]. RSM analysis was conducted by Minitab, and the significance level for all statistical analyses was 0.05.

5.1.1. Result of GA parameter setting

An optimization plot for test case 2 is shown in Fig. 11. Fig. 11 shows that the best parameter settings for population size (A), crossover rate (B), and mutation rate (C) are 100, 0.8788, and 0.1071, respectively. Fig. 12 further shows the surface plots of the two-

The best solution A of parent generation	Opened shortcut 1		Opened shortcut 2	
	Start point 1	End point 1	Start point 2	End point 2
	9	3	2	8
Solution A' by local search procedure	$\downarrow \begin{matrix} r=1 \\ s=0 \end{matrix}$		$\downarrow \begin{matrix} r=1 \\ s=0 \end{matrix}$	
	Opened shortcut 1'		Opened shortcut 2'	
	Start point 1'	End point 1'	Start point 2'	End point 2'
	8	3	3	7

Fig. 9. An example of local search.

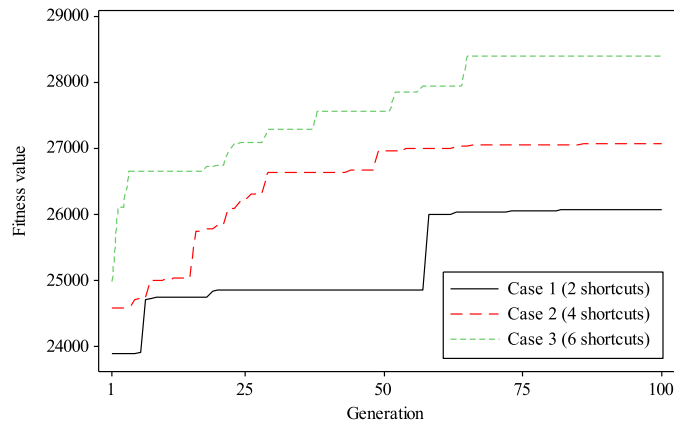


Fig. 10. Convergent condition of GA generation for three test cases.

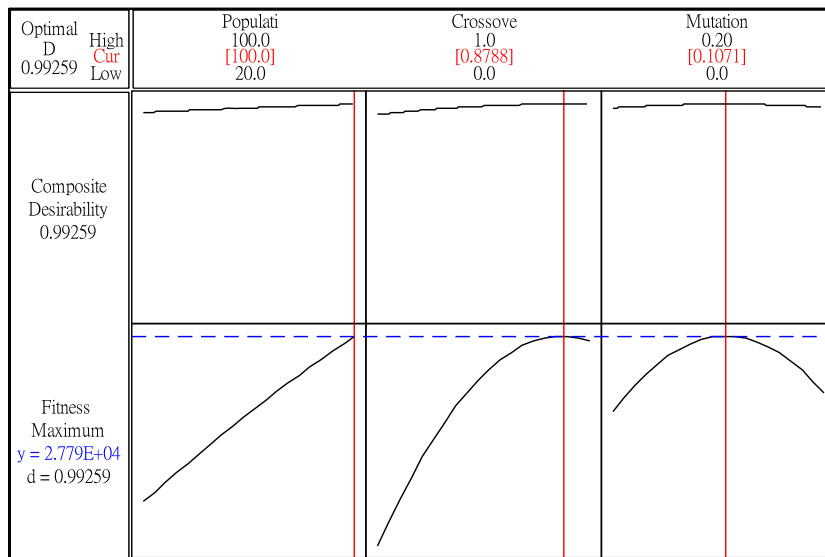


Fig. 11. Optimization plot for test case 2.

way interaction of factors A, B, and C for test case 2. For instance, when analyzing the interaction of factor A and factor B, it can be inferred that factor A should be set between 80 with 100, and factor B should be set between 0.5 with 1 to optimize the fitness value. Finally, after getting the result of the RSM, the best GA parameters are substituted into other test cases to find the optimal solution for the shortcut design.

5.1.2. Performance of GA by substituting best GA parameter

Fig. 13 shows the results of GA and LSGA for the different test cases. The LSGA yielded better system throughput by having generated a better fitness value for the GA. The throughput with the original loop design (without shortcuts) was 19,113. The result of the GA in test case 1 was the worst-performing test case, and the system throughput was enhanced by 30.2%. Test case 3 with six shortcuts enhanced the throughput of the sortation system by 48.43%. Further details, the average, standard deviation, and a 95% confidence interval, of the three cases are provided in Table 3.

Fig. 14 shows the shortcut designs of the three test cases generated by the LSGA. For each replication of test case 1 and test case 2, all five replications of the LSGA yielded the same solutions, which are seen in (a) and (b). The results show that the shortcut design is symmetrical. In case 1, shortcuts can transport parcels whose outbound conveyor is on the opposite side of the system to the other side directly instead of traveling on the loop conveyor. One of the benefits is to shorten the time to sort parcels. The other benefit is to enhance the utilization of the loop conveyor, since it saves the space of loop conveyor and allows more parcels to be loaded to the loop conveyor from the other inbound conveyor. The endpoint of the shortcut is not the nearest location to the inbound conveyor. This design can prevent parcels from being stuck at point 1 and point 7. In case 2, two shortcuts whose start/end points are points 1 and 7 are added. The benefit is also to enhance the utilization of the loop conveyor, especially the space near inbound conveyor. However, the results of each replication of test case 3 were different. The best-performing design (replication 1) is shown in Fig. 14.

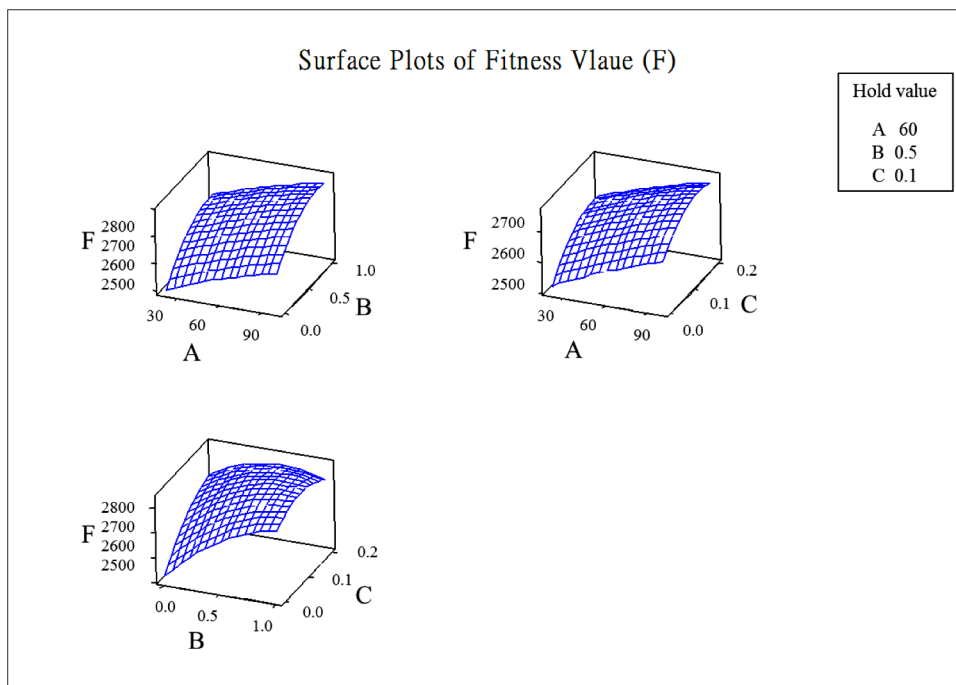


Fig. 12. Surface plot of fitness value.

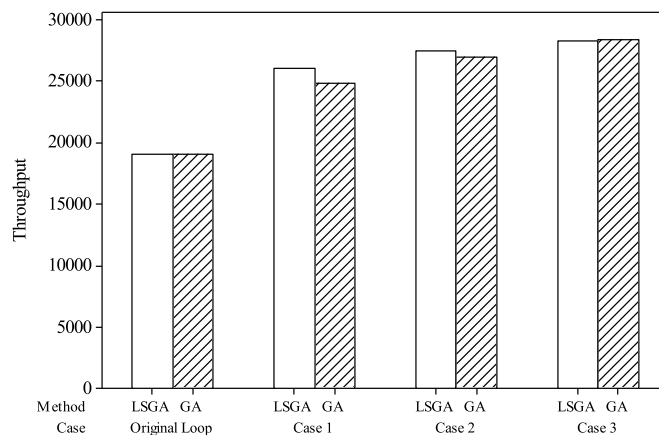


Fig. 13. The performance of GA and LSGA.

Table 3

Further details of the results of the three cases.

		LSGA	GA
Test case 1 (2 shortcuts)	Average	26,079.87	24,884.93
	Standard deviation	26	710
	Confidence interval	26,079.87 $\pm$ 22.8	24,884.93 $\pm$ 622
Test case 2 (4 shortcuts)	Average	27,550.00	27,016.00
	Standard deviation	9	711
	Confidence interval	27,550 $\pm$ 7.89	27,016 $\pm$ 623
Test case 3 (6 shortcuts)	Average	28,368.60	28,436.80
	Standard deviation	134	48
	Confidence interval	28,368.6 $\pm$ 117	28,436.8 $\pm$ 42.1

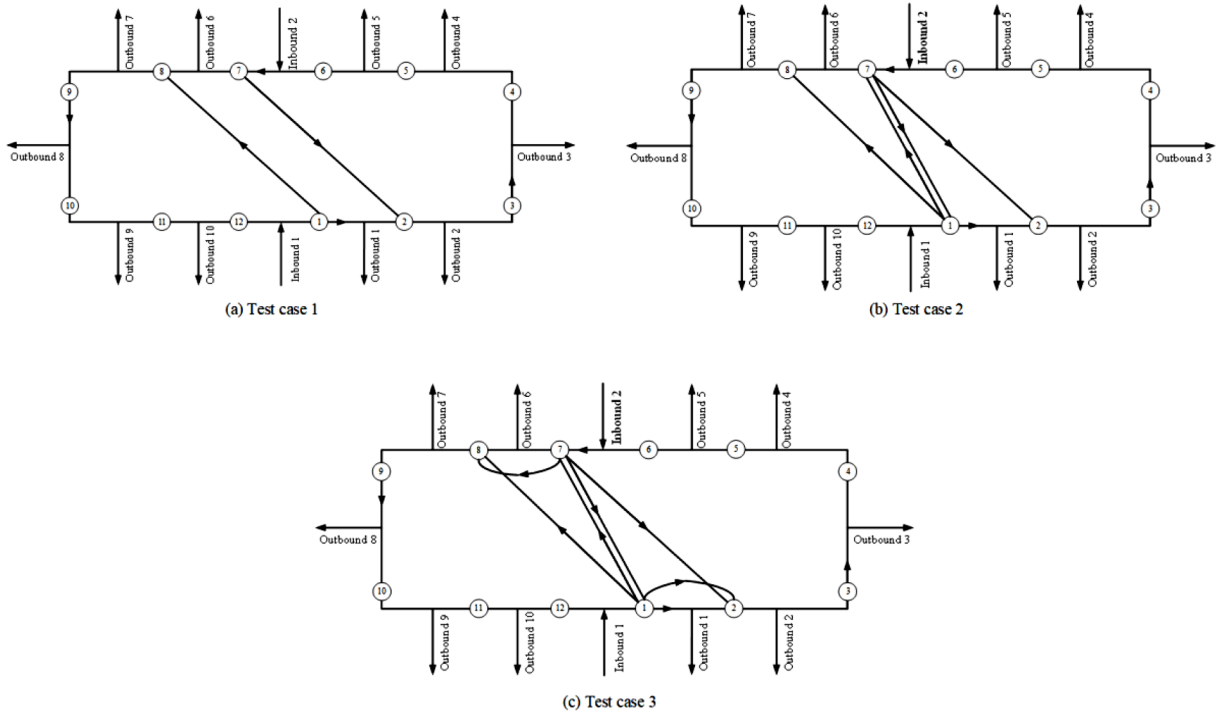


Fig. 14. The best shortcuts layout design for the three cases.

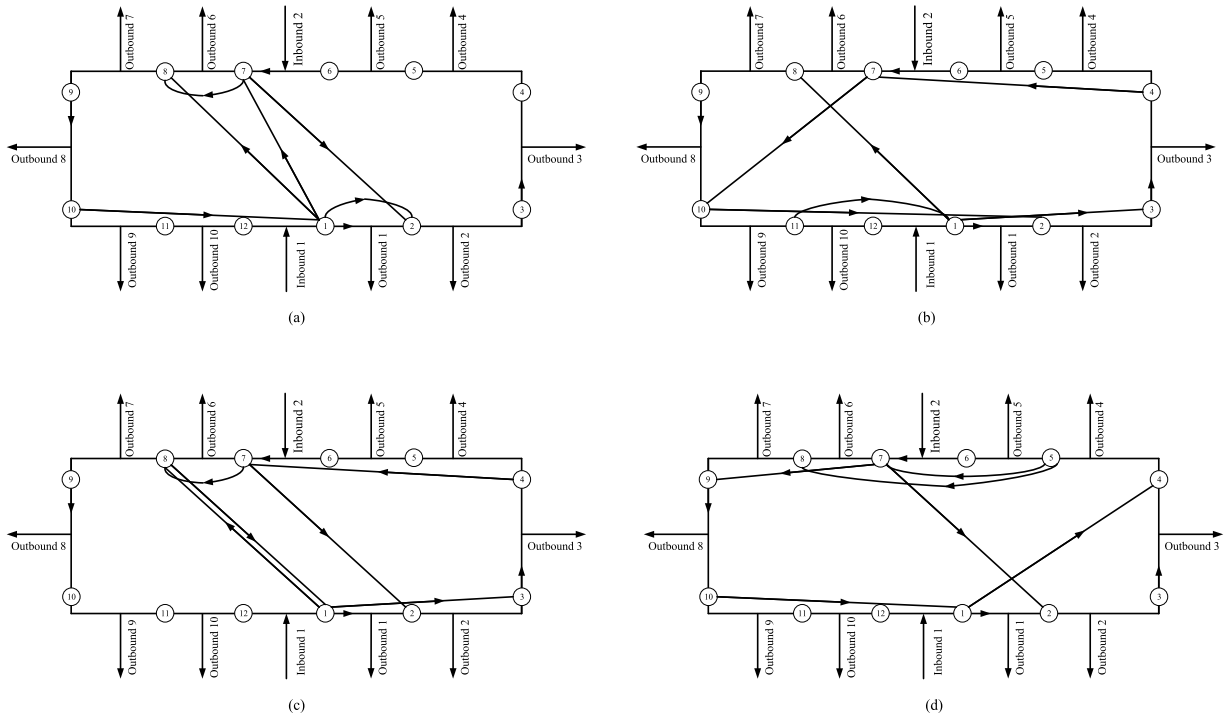


Fig. 15. Other shortcuts layout design for test case 3.

The rest of replications results of test case 3 are shown in Fig. 15. It could be found that shortcuts were primarily set at the opposite-side locations to shorten the transmission distance between two locations. Because the results of the replications are different, it can be concluded that six shortcuts are excessive for the system, so there are many designs that could reach similar optimal results.

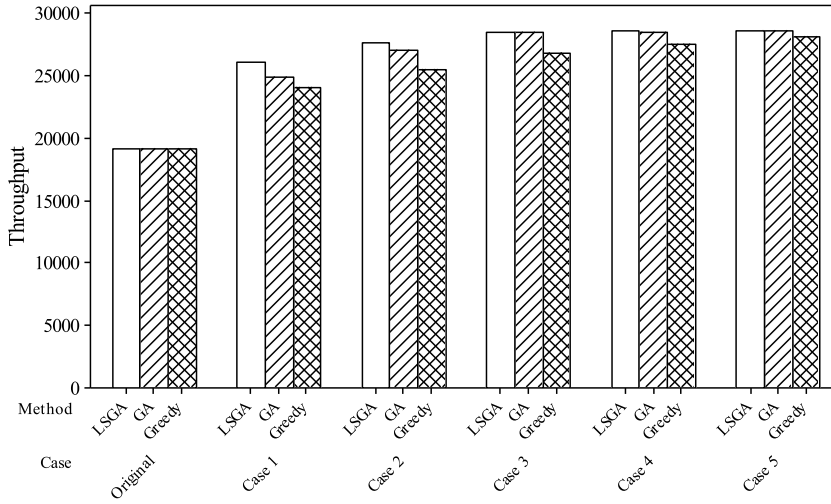


Fig. 16. Throughput value comparison between LSGA, GA and greedy algorithm.

5.2. Performance comparison for different algorithms

The three test cases with different numbers of shortcuts were used once again to evaluate the performance of the algorithm. In the past, some researchers used the greedy algorithm to solve related problems of the loop conveyor layout design in other industries. Johnson et al. [40] proposed a greedy heuristic to determine the configuration of a conveyor-based AMHS featuring turntables and crossovers to reduce WIP amount and delivery time. Hong et al. [10] used the greedy heuristic described by Johnson et al. [40] to determine the number of shortcuts in a loop conveyor layout design problem. In this study, a greedy algorithm is applied to solve the shortcut design problem too. The greedy algorithm was chosen as an alternate algorithm to evaluate the performance of the proposed GA and LSGA.

The results of three algorithms are shown in Fig. 16. For each of the test cases, the greedy algorithm achieved a worse result than those of the GA and LSGA. Generally, the LSGA is most capable of finding the best solution among the three algorithms. This also proves that the GA and LSGA proposed are capable of achieving superior results to those of previous approaches. In test case 3, the abilities of the GA and LSGA to find an optimal solution are almost the same. This specific aspect is caused by the constraint of the maximum input. The maximum input of the sortation system can be obtained by applying Eq. (5).

$$\text{Totalinputofparcelsperhour} = \frac{\text{Conveyorspeed}(m/s) \times 60(\text{sec}) \times 60(\text{min})}{\text{Parcellength}(m)} \times \text{amountofinputconveyors} \quad (5)$$

Therefore, for the situation in this study, the total input of parcels per hour is $(2 \times 60 \times 60 \div 0.5) \times 2 = 28,800$. Because the maximum of input parcels is 28,800, it is reasonable that the LSGA does not obtain a significantly better result than GA, because they both achieved results very close to the upper constraint. To validate that test case 3 is the setting with the most acceptable performance for this layout design problem, a test case 4 with eight shortcuts and test case 5 with ten shortcuts are introduced to examine whether the throughput can be significantly improved with further increase in the number of shortcuts. In addition, there are only 12 locations to be selected as the start point or endpoint of a shortcut in this case. Too many shortcuts may result in uneconomical layout. Hence, cases with more than ten shortcuts are not considered in this sortation system. Fig. 16 shows that, for the GA and LSGA, the results of test case 4 and 5 are convergent, and only the greedy algorithm's performance is enhanced, but its solution still performed worse than those of the GA and LSGA. From the perspective of throughput, cases 4 and 5 perform better results than case 3. However, the maximum input of the sortation system is almost achieved in cases 3, 4, and 5. To consider the efficiency of shortcuts, the result shows that test case 3 with six shortcuts is the most acceptable solution for the sortation system.

5.3. Discussion on computational experiments

Despite this paper focuses on using GA to solve the conveyor layout design problem with shortcuts for the real-world sortation system in Fig. 4. However, it is still necessary to carry out the experiments on other layouts to understand the applicability of the proposed method. In the real case of Fig. 4, we have considered the layout with 2 inbound conveyers and 10 outbound conveyers. To investigate the pattern of computational time, we further consider a smaller layout and a larger layout to study the applicability of the proposed method.

For the other cases, the smaller layout is with 2 inbound conveyers and 5 outbound conveyers, while the larger layout is with 4 inbound conveyers and 20 outbound conveyers. For easily comparison, we consider two shortcuts in all experiments. Compared with the original layout in the real case, the computational time of all experiments is provided in Table 4.

Table 4
Experimental results of the other cases.

Experiment	# of inbound conveyers	# of inbound conveyers	Computational time (s)
Smaller layout	2	5	8,101
Original layout	2	10	11,923
Larger layout	4	20	26,713

6. Conclusion

In this research, a simulation model was used for a loop sortation system layout design problem (especially for shortcut design), and a sortation rule was considered in the model. Furthermore, a GA for solving the proposed problem was designed, and local search was also taken into consideration in the performance comparison. Finally, an RSM was used to evaluate the performance of the GA and simulation system. After using the RSM, optimal parameters for the algorithm can be determined. The performance evaluated in the experiments was the throughput of the system. Using test cases with different numbers of shortcuts for analysis, the LSGA was found to be capable of better performance than previous algorithms.

The contribution of this research was to define and solve a loop sortation layout design problem with efficient and effective methodologies. Furthermore, the optimal design of different shortcuts was discussed. In every logistic company, it is crucial to discover the key principle of designs based on each condition to mitigate extra costs caused by frequent alteration. The topic can be extended in the following directions.

- 1 The combination of truck scheduling and sortation operation problems can be considered. Moreover, the cost of a facility with proposed layout design can be evaluated.
- 2 The condition of more-diverse parcel types or more-complex parcel frequency and distribution, which add to complexity of the problem, should be considered.
- 3 Solutions with other possible optimization algorithms, such as dynamic programming, can be obtained to evaluate the performance further.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.simpat.2020.102192](https://doi.org/10.1016/j.simpat.2020.102192).

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